SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION with OpenCV
COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 20%

Scenario-Based Questions

1. Low Contrast Image
A grayscale image appears dull with very little difference between object and background intensities.
(a) Clearly interpret the scenario and explain the cause of low contrast.
(b) Identify all key issues affecting image visibility.
(c) Apply an appropriate enhancement technique to address the problem.
(d) Justify your choice with logical reasoning.
(e) Present your explanation in a clear and structured manner.

ANS: **Low Contrast Image**
(a) Interpretation:
A low-contrast image has very little difference between the object and the background, so the image appears dull and unclear.
(b) Key Issues:
- Poor visibility of objects.
- Low brightness difference.
- Difficult to identify image details.

(c) Enhancement Technique:
Apply **Histogram Equalization** (or Contrast Stretching) to improve the image contrast.
(d) Justification:
Histogram equalization spreads the intensity values over a wider range, making objects and details more visible.

(e) Conclusion:
Using histogram equalization improves image quality by increasing contrast, making the image clearer and easier to analyze.

2. Uneven Illumination
An image captured under non-uniform lighting shows bright and dark patches.
(a) Interpret the scenario and explain the impact of uneven illumination.
(b) Identify key factors causing intensity variation.
(c) Apply a suitable enhancement method to normalize illumination.
(d) Justify your selected approach with reasoning.
(e) Provide a clear and well-structured explanation.

ANS: **Uneven Illumination**
(a) Interpretation:
Uneven illumination means the image has bright and dark regions due to non-uniform lighting, making it difficult to see all objects clearly.
(b) Key Factors:
- Uneven light source.
- Shadows.
- Different lighting conditions.

(c) Enhancement Method:
Apply **Histogram Equalization** (or Adaptive Histogram Equalization) to normalize the illumination and improve contrast.
(d) Justification:
Histogram equalization enhances the intensity distribution, reducing the effect of uneven lighting and improving image visibility.
(e) Conclusion:
The enhancement method makes the image brighter, more uniform, and easier to analyze.

3. Salt-and-Pepper Noise
A scanned document contains salt-and-pepper noise affecting readability.
(a) Explain the scenario and characteristics of this noise.
(b) Identify key issues impacting image quality.
(c) Apply an appropriate filtering technique.
(d) Justify why this method is more suitable than alternatives.
(e) Present your response clearly and logically.

ANS: **Salt-and-Pepper Noise**
(a) Scenario:
Salt-and-pepper noise appears as random black and white dots in an image, reducing its readability.
(b) Key Issues:
- Loss of image clarity.
- Reduced readability.

- Distorted image details.

(c) Filtering Technique:
Apply a **Median Filter** to remove the noise.
(d) Justification:
The median filter effectively removes salt-and-pepper noise while preserving image edges better than an averaging filter.
(e) Conclusion:
Using a median filter improves image quality by removing noise and keeping important details intact.

4. Blurred Edges
An image processed for noise removal appears overly smooth, causing edges to blur.
(a) Interpret the scenario and explain why edges are blurred.
(b) Identify key issues related to smoothing and edge loss.
(c) Apply an appropriate technique to restore edges.
(d) Justify your decision with clear reasoning.
(e) Structure your explanation clearly.

ANS: **Blurred Edges**
(a) Interpretation:
The image becomes too smooth after noise removal, causing edges and fine details to appear blurred.
(b) Key Issues:

- Loss of edge details.
- Reduced image sharpness.
- Difficulty in identifying object boundaries.

(c) Technique:
Apply **Sharpening using the Laplacian Filter** (or High-pass Filter).
(d) Justification:
The Laplacian filter enhances edges and restores image sharpness by highlighting intensity changes.
(e) Conclusion:
Sharpening improves edge clarity and makes the image clearer for analysis.

5. High-Frequency Noise
A satellite image contains fine-grained noise affecting feature visibility.
(a) Interpret the scenario and describe high-frequency noise.
(b) Identify the key issues affecting image clarity.
(c) Apply an appropriate frequency domain filtering method.
(d) Justify the choice of filter with reasoning.
(e) Present your answer in a clear and organized manner.

ANS: **High-Frequency Noise**

(a) Interpretation:
High-frequency noise appears as fine-grained random variations in the image, making features difficult to see.
(b) Key Issues:

- Reduced image clarity.
- Important features are hidden.
- Difficulty in image analysis.

(c) Filtering Method:
Apply a **Low-Pass Filter** in the frequency domain to remove high-frequency noise.
(d) Justification:
A low-pass filter removes high-frequency noise while preserving the important image information.
(e) Conclusion:
Using a low-pass filter improves image quality and makes features more visible.

Code of Conduct:

I AMALRAJ P (192524089) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightag e (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understa nding of Scenario	20%	Demonstrate s clear and complete understandin g of the scenario and context	Shows good understandin g with minor gaps	Partial understandi ng; misses key aspects	Misinterpret s or fails to understand the scenario
Identificat ion of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Applicatio n of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts

Decision Making	20%	Makes well-reasone d, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Respons e	15%	Response is clear, structured,	Generally clear with minor issues	Somewhat unclear or	Disorganize d and
		and logically presented		poorly structured	difficult to understand